Pratik Mane

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Skills

Languages: Python, R, C++, Java, JavaScript, C#, SQL, HTML, CSS
Backend & APIs: REST APIs, Node.js, JWT/OAuth, Express
Data and Cloud: Airflow, Kafka, dbt, BigQuery, Google Cloud Storage, Docker, Terraform, AWS, Databricks
Frameworks: React, Next.js, Vue.js
Databases: PostgreSQL, MySQL, MongoDB
Tools and Platforms: Git, GitHub, RStudio, Figma, Google Analytics, Power BI, Quarto

Experience

Data Engineer, *COMET Lab* Aug 2026 – Present

- Prevented wrong crime figures by validating against 2021 data and official FBI figures, catching an agency ORI code error that undercounted incidents by nearly 99% and confirming all offense categories within 3-7%
- Combined separate FBI crime data files into one clean, incident-level dataset in R (dplyr), keeping only the most serious offense per incident so researchers could analyze Indiana crime without double counting
- Built a population-coverage pipeline joining FBI agency data with U.S. Census estimates across all 92 Indiana counties, finding 62.6% average coverage and 49 counties meeting the 85% coverage threshold

Backend Developer, *The Polis Center* Jan 2026 – Aug 2026

- Removed the manual work of writing ADA-compliant alt text by building the backend that connects AI vision models to uploaded PDF, PPTX and DOCX files and generates the text automatically for screen readers
- Built JWT authentication from scratch (FastAPI, SQLAlchemy, bcrypt) so each user's documents stay private
- Built forgot-password and reset-password endpoints backed by a token table with 15-minute expiry and single-use enforcement, managing schema changes via versioned Alembic migrations on PostgreSQL
- Linked uploaded documents to their owners via a user_id foreign key and built a history endpoint returning a user's full document and processing history, enabling secure per-user data retrieval and easy access to past results
- Collaborated with the lead developer via Git branching, pull requests, and code review; diagnosed and fixed a PDF extraction bug, contributing to a three-strategy image-extraction pipeline for uploaded documents

Software Engineering Intern, *XR4RAP LLC* Jun 2025 – Aug 2025

- Built a therapeutic VR pilot for Meta Quest (C#, Unity XR Toolkit, XR Origin) for guided rehab experiences
- Integrated AI APIs for narration, music, sound effects and visuals, vetting sources for IP-safe usage
- Implemented locomotion, teleportation, and interaction systems with timed scenes for guided therapeutic exercises
- Debugged camera-tracking and UI conflicts, improving user stability and comfort in extended VR sessions

Software Engineer, *RBK Technologies Limited* Sep 2022 – Dec 2023

- Gathered requirements from business and engineering teams and turned them into responsive page structures and reusable components, acting as the technical point of contact from Figma prototype to engineering signoff
- Built and deployed a responsive Next.js, React and Headless UI website, optimizing rendering performance, accessibility and cross-browser compatibility across desktop and mobile, supporting consistent user engagement
- Set up Hostinger deployment and GA4 tracking for traffic, user-behavior and content decisions
- Cut page load times 25% with asset optimization and caching; used Git and GitHub for reviews and releases

Projects

Indiana Hospital Capacity & Strain Tracker: Built a data pipeline with CI/CD via GitHub Actions, running nine dbt data tests on every push to main, and validated results against prior-year data to catch a real discrepancy.

Ames Housing Market Intelligence: Analyzed Ames, Iowa home sales in R using hypothesis tests, regression and time series to show what drives price and the best time to buy or sell, published as a Quarto website.

AI Image Captioning: Built an AI-driven image captioning system leveraging Convolutional Neural Networks (CNN) for feature extraction and BLIP (Bootstrapping Language-Image Pretraining) for natural language caption generation.

Student Academic Performance Prediction: Built a Streamlit app in Python to explore student and school data and predict final grades, helping flag at-risk students early, with EDA, visualization and prediction pages.

Education

Indiana University Expected Dec 2026
Master of Science in Data Science

Purdue University Jan 2024 – Dec 2025
Master of Science in Computer Science